

Multi-asset market impact and order flow commonality

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Abstract

The empirical finding that market movements in stock prices may be correlated with the order flow of other stocks has led to the notion of "cross-impact" and has prompted the development of multivariate models of market impact. These models are parametrised by a matrix of impact coefficients whose off-diagonal elements are meant to capture how trades in one asset influence the price of other assets, leading to a large number of cross-impact parameters which may not be identified solely based on the covariance of returns with order flow. Moreover, empirical evidence suggests that the sign and amplitude of these cross-impact terms are unstable over time, which poses a problem for their interpretation and use.

We reexamine this empirical evidence from a causal standpoint and offer a simpler explanation for the observed correlation between the returns of an asset and the order flow imbalance of other assets, in terms of common components in order flow across stocks which may naturally arise from multi-asset trading strategies. We provide empirical evidence from order flow and price changes of NASDAQ-100 stocks to support this explanation. Our results show the main determinants of impact to be idiosyncratic order flow imbalance as well as a market order flow factor common across stocks. Additional cross-impact terms account for less than 1% of total impact. This leads to a parsimonious approach for *causal* modelling of multi-asset market impact, which does not require introducing any concept of cross-impact.

Keywords: Market impact; Cross-impact; Trade execution; Liquidity; High-frequency data; Order flow imbalance; Causal models; Principal component analysis.

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1. Introduction

1.1. Market impact and cross-impact

The notion of *cross-impact* refers to the fact that the price of a given asset may be influenced not only by its own order flow, but also by the order flow of *other* assets, thus resulting in higher execution costs when more than one security is traded at the same time. The empirical finding of positive covariance between the returns of one asset and the order flow imbalance of other assets has been hailed as evidence of cross-impact, leading to multi-asset market-impact models with off-diagonal *cross-impact coefficients* (Alfonsi et al., 2016; Benzaquen et al., 2017; Mastromatteo et al., 2017; Min et al., 2018).

In their simplest form, such models may be seen as multi-asset extensions of the Kyle (1985) model, parameterised by a matrix of impact coefficients whose off-diagonal elements are meant to capture how trades in one asset influence the price of other assets (Caballé and Krishnan, 1994). Such multi-asset impact models have been the subject of several recent theoretical and empirical studies (Andrade et al., 2008; Schied et al., 2010; Boulatov et al., 2013; Pasquariello and Vega, 2015; Alfonsi et al., 2016; Schöneborn, 2016; Benzaquen et al., 2017; Mastromatteo et al., 2017; Tsoukalas et al., 2017; Min et al., 2018; Schneider and Lillo, 2019; Tomas et al., 2020). Some models (e.g. Schied et al., 2010; Schöneborn, 2016) *assume* the existence of cross-impact without detailing the underlying causal mechanism that might give rise to this phenomenon. Pasquariello and Vega (2015) present a model of sequential trading in which market makers' quote revisions give rise to cross-impact.

Despite the wide use of such models, the theoretical and empirical evidence underlying their justification is not entirely convincing for a number of reasons. The mere presence of positive covariation between the returns of an asset and the order flow of another asset is not sufficient to provide a causal justification for the existence of cross-impact as a separate phenomenon from, for example, correlation between the order flow of the two assets. Common components in order flow - which are well documented empirically (Hasbrouck and Seppi, 2001) naturally arise from cross-asset trading (Bernhardt and Taub, 2008), index trading (Harford and Kaul, 2005) and portfolio rebalancing (Cont and Wagalath, 2016) may lead to similar effects without the need to invoke cross-impact.

Despite the wide use of such models, the theoretical and empirical evidence underlying their justification is not entirely convincing for a number of reasons. The mere presence of positive covariation between the returns of an asset and the order flow of another asset might not be sufficient to provide a causal justification for the existence of cross impact as a separate phenomenon from, for example, correlation between the order flow of the two assets. Common components in order flow - which are well documented empirically (Hasbrouck and Seppi, 2001) - naturally arise from cross-asset trading (Bernhardt and Taub, 2008), index trading (Harford and Kaul, 2005) and portfolio rebalancing (Cont and Wagalath, 2016) may lead to similar effects without the need to invoke cross impact.

In this paper we investigate these questions using a *causal* modelling approach (Haavelmo, 1943, 1944; Heckman and Pinto, 2015; Pearl, 2009), carefully distinguishing measures of association - such as correlations - from causal relations, to arrive at a causal model for market impact in a multi-asset setting.

In particular, we investigate whether there is any empirical evidence for the concept of cross-impact or whether a simpler and more parsimonious explanation may be given for the observed correlations among returns and order flows across assets without invoking this notion.

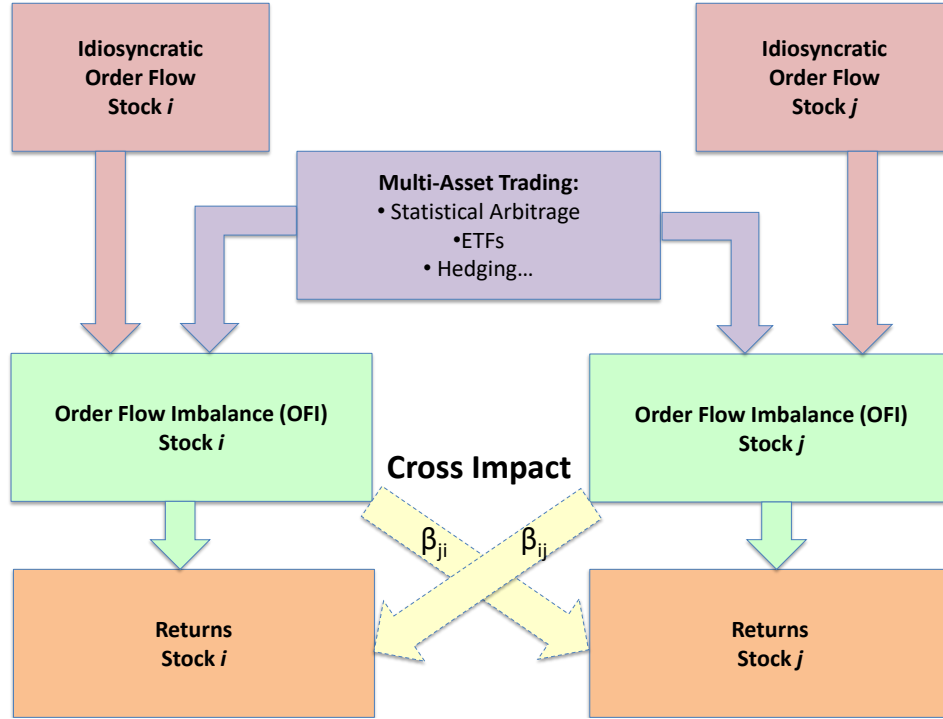


Fig. 1. Causal representation of multi-asset impact models. The returns of each asset are influenced by the asset's own order flow imbalance, which is the superposition of an idiosyncratic order flow and of a *common component* across assets. This common component leads to correlations in order flow imbalance across assets. Cross-impact models posit the existence of an additional direct influence of the order flow imbalance of an asset on the returns of other assets, represented by the diagonal arrows in the causal diagram.

1.2. Contribution: causal analysis and modelling of multi-asset market impact

In the present study, we reexamine the empirical evidence from equity markets and provide a different perspective on the notion of cross-impact.

Starting from a linear model for market impact, we show that cross-impact is not needed to explain the positive covariation between the returns of an asset and the order flow of other assets. Instead, we provide evidence that this covariation simply arises from the correlation of order flow imbalances across assets. A significant proportion of the demand and supply for assets originates in portfolio rebalancing strategies or trading strategies -index or ETF trading, pairs trading, statistical arbitrage strategies and hedging strategies- which involve positions in several assets, thereby generating correlations in order flow.

We investigate these questions through the angle of causal modelling (Haavelmo, 1943, 1944; Heckman and Pinto, 2015; Pearl, 2009), which emphasises causal relations among variables, testable through conditioning.

Figure 1 shows the causal chain of relations for two such assets i and j : market participants trade in stocks i and j at the same time, which leads to correlation between the OFIs of the two stocks. The return of each asset is determined by the order flow imbalance, which is the superposition of a *common component* and the idiosyncratic order flow in each security. The presence of this common component leads to correlations in order flow imbalance across assets, and results in correlations between the returns of an asset and order flow of other assets.

Cross-impact models insert an additional mechanism, represented in Figure 1 as a pair of diagonal arrows,

through which the order flow of one asset would somehow influence the prices of another asset, bypassing the supply/demand mechanism usually assumed to underpin price formation.

The causal graph in Figure 1 suggests that without conditioning on the common component in the order flow, one cannot distinguish between cross-impact and order flow correlation. In Section 2.2, we present examples that illustrate this *identifiability* problem, and demonstrate that one can overcome this problem by simply estimating the covariance of the order flow imbalances.

The presence of causal relations, shown as arrows in Figure 1, may be tested by conditioning on the variable hypothesised to contribute to an observed effect. We use this approach to test for the significance and amplitude of cross-impact channels by conditioning on the common factor in order flow imbalance. Our results, in Section 3.3, show that once a stock's own order flow and the commonality in order flow has been taken into account, the additional explanatory power of all other stocks' order flow is less than 0.5 percentage points.

Finally, in section 3.5 we show that, in addition to being small in magnitude, cross-impact coefficients are unstable and change sign randomly over time, which poses a problem for their interpretation and use.

Our results show the main determinants of impact to be idiosyncratic order flow imbalance as well as a market order flow factor common across stocks. This leads to a parsimonious approach for *causal* modelling of multi-asset market impact, which does not require introducing any concept of cross-impact.

Our findings have important implications for optimal trade execution and transaction costs analysis. They show that accounting for commonality in order flow seems to be a more effective way to build multi-asset market impact models than introducing cross-impact coefficients whose statistical significance and interpretation are not clear.

1.3. Relation with previous literature

The extension of the model in Kyle (1985) to multiple assets led, almost inevitably, to trying to understand and model how prices in an asset may be influenced by trades in other assets. Caballé and Krishnan (1994) developed a model that relates prices and order flows of different assets and derive the consequences for covariance structure of these variables. Pasquariello and Vega (2015) present a multi-asset sequential trade model in which cross-impact arises, in equilibrium, from the strategic cross-trading of informed speculators and the market makers' subsequent cross inference based on the speculators' order flow and present empirical evidence that cross-impact is economically and statistically significant, and often negative.

Boulatov et al. (2013) present a multi-period model of strategic trading in which cross-impact arises as a results of market makers updating their quotes on an asset based on the order flow of other assets. Using institutional investor orders on the NYSE, these authors find statistical evidence of a lead-lag correlation between the returns of a stock and the informed order flow on another stock.

Andrade et al. (2008) propose a model in which the magnitude of cross-impact depends on the correlation among stocks' underlying cash flows, and their relative attractiveness as hedging instruments for liquidity providers, and provide empirical evidence from the Taiwan Stock Exchange. Our results, although at intraday frequencies and using different data sources, are consistent with these findings.

Cross-impact has also been considered in the context of optimal execution in multivariate extensions of Almgren and Chriss (2001). The focus in these studies is the formulation of strategies for optimal portfolio liquidation in the presence of volatility risk. For example, Schied et al. (2010) and Schöneborn (2016) use this framework to study optimal execution strategies for investors with constant absolute risk aversion, and for a more general absolute risk aversion setting, respectively. However, the authors *assume* cross-impact,

rather than detailing the mechanism through which it might arise. Tsoukalas et al. (2017) develop a tractable order book model to capture liquidity supply and demand dynamics in a multi-asset setting and show that even when the goal is to trade a single asset, optimal execution may involve transitory trades in other assets, leading to cross-impact. However, no empirical evidence is provided to support this model, which does not distinguish cross-impact from order flow correlation, an important point which we address in this paper.

Other models of cross-impact in trade execution include Mastromatteo et al. (2017), Min et al. (2018) and Tomas et al. (2020). Finally, the notion of cross-impact has also been studied from the perspective of absence of dynamic arbitrage by Alfonsi et al. (2016) and Schneider and Lillo (2019). In these papers, cross-impact is, once again, assumed rather than derived.

Commonalities in liquidity and order flow have been widely documented in the literature (Chordia et al., 2000; Hasbrouck and Seppi, 2001; Heston et al., 2010; Pástor and Stambaugh, 2004) and cannot be dismissed as one of the main sources of positive covariation between the returns of a stock and the order flow of other stocks. Harford and Kaul (2005) document common factors in order flow arising from index trading; Bernhardt and Taub (2008) study order flow correlations arise from speculative cross-asset trading. Cont and Wagalath (2016) study endogenous correlations in order flow and returns arising from portfolio rebalancing of institutional investors. Cont and Wagalath (2013) show how liquidation of large positions by institutional investors may lead to correlated order flow and endogenous correlations in price movements. All these mechanisms lead to significant correlations between asset returns and order flow in other assets without introducing the concept of cross-impact. We build on these ideas and show that they lead to a plausible and parsimonious explanation for the positive covariance of returns and order flow imbalance across different assets.

Outline The remainder of the paper is organised as follows. Section 2 describes a linear model of market impact from trading and discusses the issue of identifiability of the cross-impact coefficients through an example. Section 3 presents empirical evidence for our causal model, obtained from the analysis of order flow, trades and quotes for NASDAQ stocks. Section 4 discusses the implications of our findings for the modelling of market impact.

2. Order flow correlation and cross-impact

2.1. A linear model of market impact

Linear models of market impact, going back to Kyle (1985), are based on a linear relation between price movements (or returns) and aggregate net order flow for an asset.

Cont et al. (2014) provide empirical evidence for such a linear model of market impact at intraday frequencies using limit order book data, using the concept of *order flow imbalance* (OFI):

$$R_{i,t} = \mu_i + \beta_i \text{OFI}_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where $R_{i,t}$ is the return of asset i , $\text{OFI}_{i,t}$ is the order flow imbalance for asset i , defined as the net order flow during the time interval $[t - \Delta t, t]$, and $\varepsilon_{i,t}$ is a noise term summarising influences of other factors (e.g., deeper levels of the order book, etc.).

The multi-asset extension of (1) to N assets may be written in matrix form as:

$$\mathbf{R}_t = \boldsymbol{\mu} + \boldsymbol{\beta} \mathbf{OFI}_t + \boldsymbol{\varepsilon}_t, \quad (2)$$

where $\boldsymbol{\mu}$, $\mathbf{R}_t$, and $\mathbf{OFI}_t$ are respectively $N \times 1$ vectors of constants, returns and order flow imbalances:

$$\boldsymbol{\mu} \equiv \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_N \end{bmatrix}, \quad \mathbf{R}_t \equiv \begin{bmatrix} R_{1,t} \\ R_{2,t} \\ \vdots \\ R_{N,t} \end{bmatrix}, \quad \mathbf{OFI}_t \equiv \begin{bmatrix} \text{OFI}_{1,t} \\ \text{OFI}_{2,t} \\ \vdots \\ \text{OFI}_{N,t} \end{bmatrix},$$

$\boldsymbol{\varepsilon}_t$ is an $N \times 1$ vector of uncorrelated error terms:

$$E(\boldsymbol{\varepsilon}_t) = 0, \quad \text{cov}(\boldsymbol{\varepsilon}_t) = \boldsymbol{\Sigma}, \quad \text{cov}(\boldsymbol{\varepsilon}_t, \mathbf{OFI}_t) = 0, \quad (3)$$

and $\boldsymbol{\beta}$ is an $N \times N$ matrix of market impact coefficients:

$$\boldsymbol{\beta} \equiv \begin{bmatrix} \beta_{11} & \beta_{12} & \dots \\ \vdots & \ddots & \\ \beta_{N1} & & \beta_{NN} \end{bmatrix}, \quad (4)$$

whose off-diagonal terms represent 'cross-impact'. Let $\boldsymbol{\mathcal{C}}$ and $\boldsymbol{\mathcal{K}}$ be the covariance matrices of returns and OFI:

$$\boldsymbol{\mathcal{C}} \equiv E[(\mathbf{R}_t - \boldsymbol{\mu}_R)(\mathbf{R}_t - \boldsymbol{\mu}_R)^T], \quad (5)$$

$$\boldsymbol{\mathcal{K}} \equiv E[(\mathbf{OFI}_t - \boldsymbol{\mu}_{\text{OFI}})(\mathbf{OFI}_t - \boldsymbol{\mu}_{\text{OFI}})^T], \quad (6)$$

where $\boldsymbol{\mu}_R$ is an $N \times 1$ vector of mean returns and $\boldsymbol{\mu}_{\text{OFI}}$ is an $N \times 1$ vector of mean OFIs. We can also define covariance matrix between returns and OFIs as:

$$\boldsymbol{\mathcal{M}} = \text{cov}(\mathbf{R}_t, \mathbf{OFI}_t) = E[(\mathbf{R}_t - \boldsymbol{\mu}_R)(\mathbf{OFI}_t - \boldsymbol{\mu}_{\text{OFI}})^T] = \boldsymbol{\beta} \boldsymbol{\mathcal{K}} \quad (7)$$

In (7), the diagonal terms represent the covariance between the returns of asset i and its own order flow imbalance, while the off-diagonal terms represent the covariance between the returns of asset i and the order flow imbalance of assets $j \neq i$.

2.2. Are cross-impact coefficients identifiable?

The covariance (7) of order flow imbalance with asset returns has been the subject of several empirical studies for different asset classes (Mastromatteo et al., 2017; Tomas et al., 2020). These studies find positive off-diagonal elements in $\boldsymbol{\mathcal{M}}$, and have proposed extensions of univariate impact models to multiple assets with off-diagonal, *cross-impact coefficients* in (4). But the presence of off-diagonal terms in the matrix $\boldsymbol{\mathcal{M}}$ may arise from a much simpler and more intuitive mechanism, without invoking the notion of cross-impact: namely, the fact that OFIs of different assets are correlated.

That order flow across different assets may be correlated is an intuitive consequence of the widespread use of multi-asset trading strategies. Market participants often adopt trading strategies (such as pairs trading, index arbitrage, and other statistical arbitrage strategies) that involve more than one asset. This is more apparent for some asset classes, such as fixed-income and commodities, where there are clear theoretical reasons that underpin the simultaneous trading of assets with different maturities. As shown in the causal diagram depicted in Figure 1, this leads to correlation among returns and OFIs of different assets, without any need to invoke cross-impact.

The following simple example shows that off-diagonal terms in $\mathcal{M}$ may arise from two different sources: correlation in order flow *or* off-diagonal elements in the cross-impact matrix β , leading to an identifiability problem for such cross-impact coefficients. Consider the model (2) with two assets ($N = 2$):

$$R_{1,t} = \beta_{11}\text{OFI}_{1,t} + \beta_{12}\text{OFI}_{2,t} + \varepsilon_{1,t}, \quad (8)$$

$$R_{2,t} = \beta_{22}\text{OFI}_{2,t} + \beta_{21}\text{OFI}_{1,t} + \varepsilon_{2,t}, \quad (9)$$

where we have dropped the constant terms μ_1 and μ_2 for simplicity. Assume that the covariance matrices of returns, OFIs, and errors terms are given, respectively, by:

$$\mathcal{C} = \begin{bmatrix} c_1^2 & \rho_c c_1 c_2 \\ \rho_c c_1 c_2 & c_2^2 \end{bmatrix}, \quad \mathcal{K} = \begin{bmatrix} k_1^2 & \rho_k k_1 k_2 \\ \rho_k k_1 k_2 & k_2^2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}, \quad (10)$$

so that the returns are correlated, with correlation coefficient ρ_c , OFIs are correlated with correlation coefficient ρ_k , and the error terms are uncorrelated. Under these assumptions, (7) yields:

$$\mathcal{M} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = \begin{bmatrix} \beta_{11}k_1^2 + \beta_{12}\rho_k k_1 k_2 & \beta_{11}\rho_k k_1 k_2 + \beta_{12}k_2^2 \\ \beta_{22}\rho_k k_1 k_2 + \beta_{21}k_1^2 & \beta_{22}k_2^2 + \beta_{21}\rho_k k_1 k_2 \end{bmatrix}. \quad (11)$$

We can see from (11) how the off-diagonal elements, m_{12} and m_{21} , are functions of β_{12} and β_{21} , and $\rho_k = \text{corr}(\text{OFI}_1, \text{OFI}_2)$. So from the knowledge of $\mathcal{M}$, one cannot draw any conclusions with respect to cross-impact. Indeed, different combinations of ρ_k and β_{ij} lead to the same off-diagonal elements in $\mathcal{M}$, including combinations where β_{12} and β_{21} are equal to zero.

The identification problem becomes clear when we solve (11) for β_{12} and β_{21} :

$$\beta_{12} = m_{12} \left[\frac{1}{k_2^2(1 - \rho_k^2)} \right] - m_{11} \left[\frac{\rho_k}{k_1 k_2(1 - \rho_k^2)} \right], \quad (12)$$

$$\beta_{21} = m_{21} \left[\frac{1}{k_1^2(1 - \rho_k^2)} \right] - m_{22} \left[\frac{\rho_k}{k_1 k_2(1 - \rho_k^2)} \right]. \quad (13)$$

The coefficients β_{12} and β_{21} can only be identified by estimating or assuming a value for the OFI correlation ρ_k . If one assumes the OFI correlation to zero then (12)-(13) collapses to:

$$\beta_{12} = \frac{m_{12}}{k_2^2}, \quad \beta_{21} = \frac{m_{21}}{k_1^2}. \quad (14)$$

In this case, β_{12} and β_{21} are identifiable and the off-diagonal elements of $\mathcal{M}$ are solely due to cross-impact. Two numerical examples help illustrate this identification problem.

Example 2.1. Consider the case where order flows are uncorrelated ($\rho_k = 0$) and normalised to have unit

variance, with market impact coefficients given by:

$$\boldsymbol{\kappa}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{\beta}_0 = \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 0.6 \end{bmatrix}. \quad (15)$$

Assuming uncorrelated error terms, the covariance matrix of returns and OFIs is given by:

$$\boldsymbol{\mathcal{M}}_0 = \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 0.6 \end{bmatrix}. \quad (16)$$

Now, consider the case where OFIs are correlated, with correlation coefficient $\rho_k = 0.5$, and the matrix of market impact coefficients is the same as in (15), but with the off-diagonal elements set to zero:

$$\boldsymbol{\kappa}_1 = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \quad \boldsymbol{\beta}_1 = \begin{bmatrix} 0.6 & 0 \\ 0 & 0.6 \end{bmatrix}, \quad (17)$$

which leads to the following covariance matrix between returns and OFIs:

$$\boldsymbol{\mathcal{M}}_1 = \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 0.6 \end{bmatrix}. \quad (18)$$

We observe that $\boldsymbol{\mathcal{M}}_1 = \boldsymbol{\mathcal{M}}_0$, yet they arise from two very different mechanisms: in the first case, order flows are uncorrelated across assets and cross-asset correlations between order flow and returns arise from cross-impact; in the second case, order flows are correlated, and there is no cross-impact mechanism. This examples shows that the mere observation of positive or non-zero off-diagonal terms in (16) is not sufficient to draw any conclusions regarding cross-impact.

Example 2.2. Consider now the case where $\rho_k = 0.2$, with OFIs normalised to have unit variance, with impact coefficients given by:

$$\boldsymbol{\kappa}_2 = \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix}, \quad \boldsymbol{\beta}_2 = \begin{bmatrix} 0.6 & 0.5 \\ 0.5 & 0.6 \end{bmatrix}. \quad (19)$$

The covariance between of returns and OFI is then given by:

$$\boldsymbol{\mathcal{M}}_2 = \begin{bmatrix} 0.7 & 0.62 \\ 0.62 & 0.7 \end{bmatrix}. \quad (20)$$

Now, consider the case where OFIs are correlated, with correlation coefficient $\rho_k = 0.6$, and the matrix of market impact coefficients market impact coefficients are: $\beta_{11} = \beta_{22} = 0.5125$, and $\beta_{12} = \beta_{21} = 0.3125$. That is to say:

$$\boldsymbol{\kappa}_3 = \begin{bmatrix} 1 & 0.6 \\ 0.6 & 1 \end{bmatrix}, \quad \boldsymbol{\beta}_3 = \begin{bmatrix} 0.5125 & 0.3125 \\ 0.3125 & 0.5125 \end{bmatrix}. \quad (21)$$

Assuming, once again, that the error terms are uncorrelated, the covariance matrix of returns and OFIs is given by:

$$\boldsymbol{\mathcal{M}}_3 = \begin{bmatrix} 0.7 & 0.62 \\ 0.62 & 0.7 \end{bmatrix}. \quad (22)$$

Again, we observe $\mathcal{M}_3 = \mathcal{M}_2$ yet these covariances result from two very different mechanisms: in the first case, there was low correlation in the order flow and high cross-impact; in the second case, there is high correlation in the order flow, but low cross-impact. As in Example 2.1, this example shows that cross-impact coefficients may not be identified solely from the covariance of returns with order flow.

3. Empirical analysis of returns and order flow imbalances

3.1. Data sources and quantities of interest

We use intraday order book data from LOBSTER,¹ which provides detailed tick-by-tick limit order book data for NASDAQ stocks. LOBSTER contains all limit order submissions, cancellations and executions on each trading day and all events are time-stamped with millisecond precision (Bibinger et al., 2017), reconstructed from NASDAQ's historical TotalView-ITCH data (Huang and Polak, 2011). The LOBSTER *message* files contain the order type, order ID, size and the price direction (buy/sell). The LOBSTER *orderbook* files contain the bid and ask prices and quantities. In our study, we use "Level I" data from LOBSTER; these are all order-book events at the best bid and ask prices (Huang and Polak, 2011).

From the LOBSTER database, we select all stocks belonging to the NASDAQ-100 Index. Our empirical analysis focuses on the period between 01-Jan-2008 and 30-Jun-2010, and after filtering out the stocks for which the dataset does not span the whole sample period, we end up with 67 stocks.

For each stock i , we estimate the daily volatility σ_i as the square root of the daily realised volatility estimator from 5-minute returns (Andersen et al., 2003).

In our analysis we will focus on $\Delta t = 1$ minute intervals. For each one-minute window $[t - \Delta t, t]$, and for each stock i , we compute:

- The return $R_{i,t}$, defined as the percentage change of the mid price between $t - \Delta t$ and t ;
- The net (buy) order flow $V_{i,t}^b$ at the best bid during $[t - \Delta t, t]$:

$$V_{i,t}^b \equiv L_{i,t}^b - C_{i,t}^b - M_{i,t}^b,$$

where $L_{i,t}^b$ and $C_{i,t}^b$ denote respectively the volume of new limit buy orders and cancellations at the current best bid and $M_{i,t}^b$ denotes the volume of marketable sell orders flowing to the current best bid during $[t - \Delta t, t]$;

- The net (sell) order flow $V_{i,t}^a$ at the best ask during $[t - \Delta t, t]$:

$$V_{i,t}^a \equiv L_{i,t}^a - C_{i,t}^a - M_{i,t}^a,$$

where $L_{i,t}^a$ and $C_{i,t}^a$ denote respectively the volume of new limit sell orders and cancellations at the current best ask and $M_{i,t}^a$ denotes the volume of marketable buy orders flowing to the current best ask during $[t - \Delta t, t]$.

The *order flow imbalance* $\text{OFI}_{i,t}$ for the interval $[t - \Delta t, t]$ is defined as (Cont et al., 2014):

$$\text{OFI}_{i,t} = V_{i,t}^b - V_{i,t}^a. \quad (23)$$

¹<https://lobsterdata.com>.

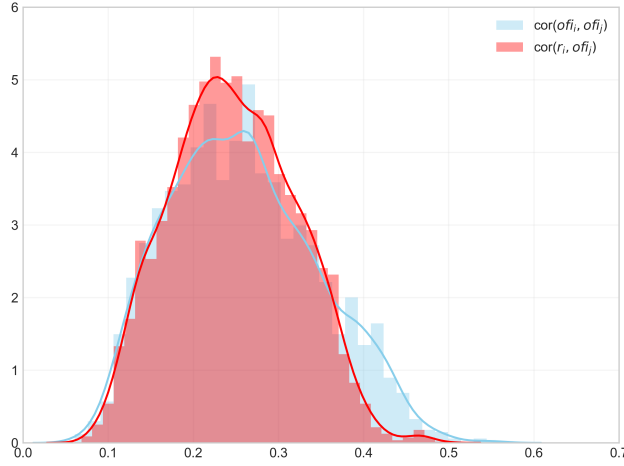


Fig. 2. Comparison of the distribution of correlations between 1-minute $\text{OFI}_{i,t}$ and $\text{OFI}_{j,t}, \forall i \neq j$ (blue) with the distribution of correlations between 1-minute $r_{i,t}$ and $\text{OFI}_{j,t}, \forall i \neq j$ (red) (Jan-2008: Jun-2010).

To account for intraday seasonality patterns in volume and order flow, we define $AV_{i,t}$ as the average total order flow $V_{i,t}^b + V_{i,t}^a$ at the best bid and ask over the (one-minute) time bucket $[t - \Delta t, t]$ and use this variable to define an average intraday profile of order flow.

In order to make our results easier to interpret, we normalise $R_{i,t}$ and $\text{OFI}_{i,t}$ as in Obizhaeva (2012). Denoting by σ_i the volatility of asset i we define:

$$r_{i,t} = \frac{R_{i,t}}{\sigma_i \sqrt{\Delta t}}, \quad \text{ofi}_{i,t} = \frac{\text{OFI}_{i,t}}{AV_{i,t}} \quad (24)$$

Figure 2 shows the distribution of the correlation coefficients between $\text{ofi}_{i,t}$ and $\text{ofi}_{j,t}, j = 1 \dots N$ (blue), and the distribution of the correlation coefficients between $r_{i,t}$ and $\text{ofi}_{j,t} \quad \forall i, j = 1 \dots N, i \neq j$ (red). We observe that the correlation coefficients between the returns and OFI of different stocks seem to follow the same distribution of the correlation coefficients between OFIs. The average pairwise correlation of OFI across stocks is 0.26 while the average correlation between returns and OFI is 0.25. The distribution of the correlation between returns and OFI is slightly more concentrated around the mean, with a standard deviation of 0.07 versus 0.09 for OFI correlations.

Figure 3 compares the correlation between returns and OFI and the correlation between OFIs for the same pair of stocks. For most pairs of stocks these values are close to being equal, suggesting that the correlation between order flow and returns may arise from correlation of order flows themselves. To test this assumption we now proceed to do a more detailed analysis based on a comparison of nested regression models and principal component analysis.

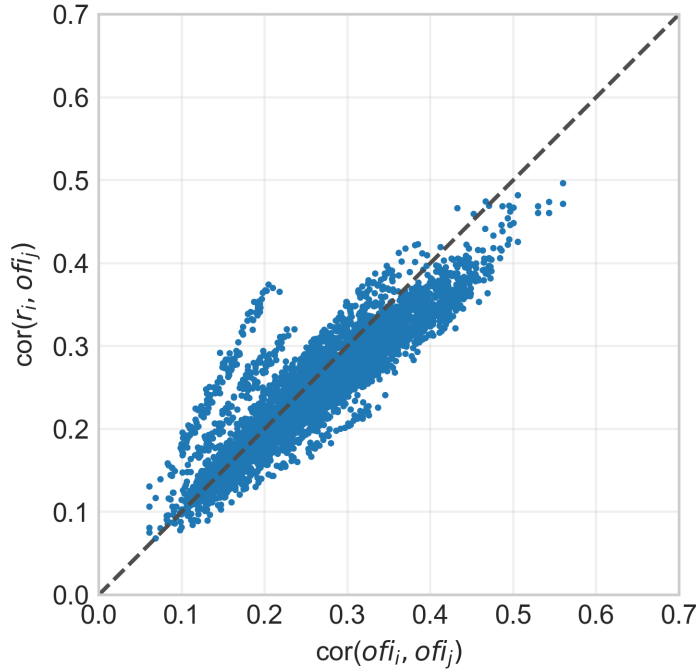


Fig. 3. Comparison of pairwise correlations between 1-minute $\text{OFI}_{i,t}$ and $\text{OFI}_{j,t}, \forall i \neq j$ with pairwise correlations between 1-minute $r_{i,t}$ and $\text{OFI}_{j,t}, \forall i \neq j$ for the same equity pairs (Jan-2008: Jun-2010). The dashed line corresponds to equality.

3.2. Regression analysis

We estimate the linear regression model (1) using ordinary least squares:

$$\mathbf{r}_t = \boldsymbol{\mu} + \boldsymbol{\beta} \text{ofi}_t + \boldsymbol{\varepsilon}_t. \quad (25)$$

and examine whether off-diagonal terms $\beta_{ij}, i \neq j$ have significant explanatory power.

Figure 4 shows a heat map of the price-impact matrix $\boldsymbol{\beta}$, estimated from (25). It is immediately clear from this heat map that the off-diagonal terms, representing the impact of other stocks order flows on a stock's returns, are much smaller in magnitude than the diagonal terms, which represent the impact of a stock's own order flow on its returns. Figure 5 further illustrates this point by presenting the relative frequency distributions of the diagonal (idiosyncratic) impact coefficients β_{ii} (blue) and of the off-diagonal cross-impact coefficients β_{ij} (red). The mean diagonal coefficient β_{ii} is 2.64 with a standard deviation of 0.78, while the mean off-diagonal cross-impact coefficient β_{ij} is only 0.032 with a standard deviation of 0.06. This means that the off-diagonal terms are only about 1.2% in magnitude when compared with the diagonal terms. Further, Figure 5 shows that whilst most off-diagonal terms are positive (76.91%), a number of cross-impact coefficients are negative (23.09%), indicating cross-impact in the opposite direction to what is suggested by many cross-impact models.

Given these results, we test the hypothesis that the off-diagonal terms are equal to zero, using Newey-West heteroskedasticity-consistent estimators for standard errors. Out of 4,422 off-diagonal terms, 1846 (41.3%)

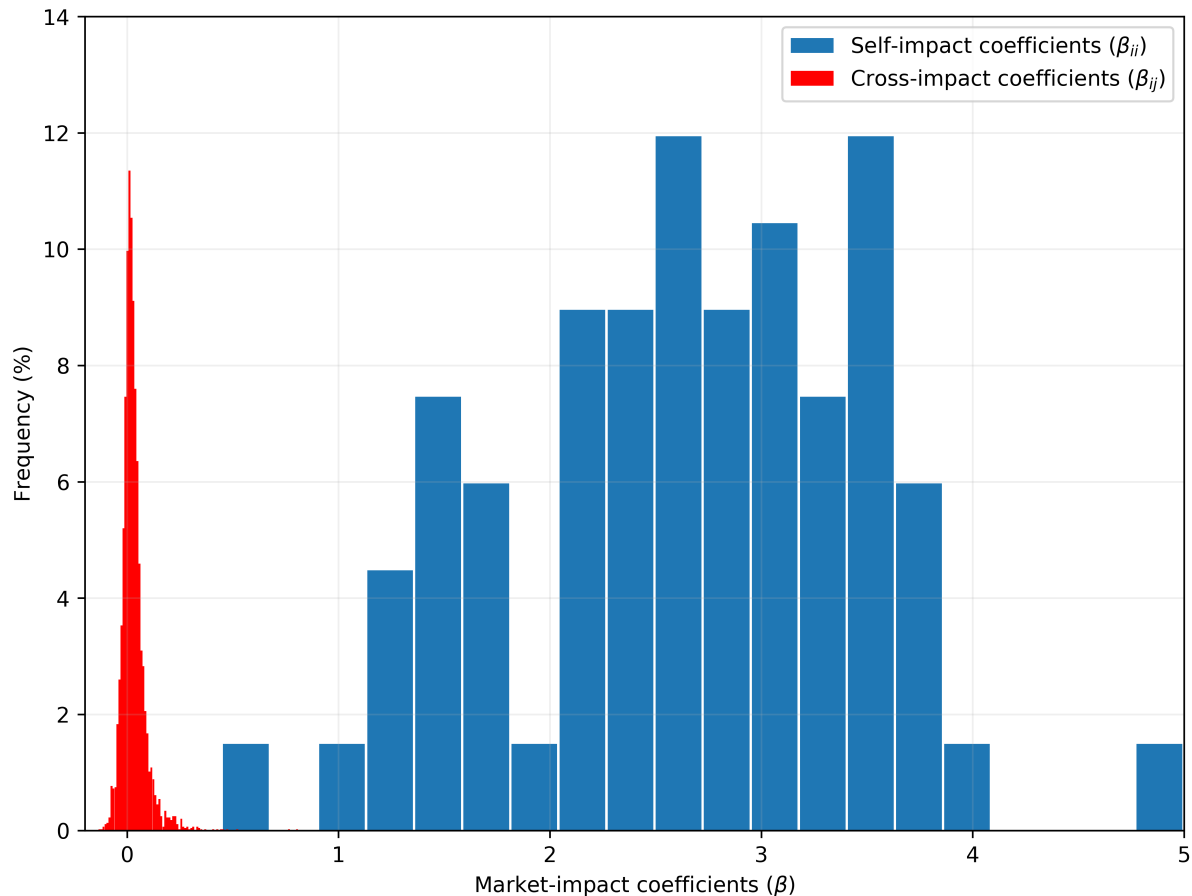


Fig. 5. Distribution of the diagonal (idiosyncratic) impact coefficients β_{ii} (blue) and off-diagonal ("cross-impact") coefficients β_{ij} (red) estimated from (25) (Jan-2008: Jun-2010).

3.3. Common components in order flow imbalance

As shown in Section 2.2, failing to account for the presence of common factors in order flow leads to an identification problem and makes it difficult to draw conclusions regarding the presence of cross-impact as a phenomenon distinct from correlation in order flow.

We account for the commonality in order flow by performing a principal component analysis (PCA) on the correlation matrices of returns and OFIs. We adjust for the correlation in order flow using the first principal component of OFI, then test for the statistical significance of any residual cross-impact effect.

Table 1 reports the results of this PCA for the returns and order flow imbalances of the 67 stocks under consideration. For each of these two variables, we report the eigenvalues, the variance explained by each eigenvalue and the cumulative variance explained by the sum of the current and all previous eigenvalues. The sum of eigenvalues is equal to $N = 67$.

From Table 1, we see that the first eigenvalue of returns is 21.44, implying that $21.44/67 = 32.00\%$ of the total variation in 1-minute returns can be explained by a single common factor. These results suggest a stronger commonality in intraday returns than reported in previous studies: Hasbrouck and Seppi (2001) found that 21% of total variation in 15-minute returns on 30 Dow Jones stocks was accounted for by the

	Returns			Order Flow Imbalance		
Eigenvalue	(1)	(2)	(3)	(1)	(2)	(3)
Value	21.44	1.70	1.48	18.94	1.29	1.21
Explained Variance	32.00%	2.54%	2.21%	28.27%	1.92%	1.80%
Cum. Explained Variance	32.00%	34.54%	36.75%	28.27%	30.19%	32.00%

Table 1: Principal component analysis of returns and order flow imbalance (Jan-2008: Jun-2010). The table shows the first three eigenvalues, the variance explained by each eigenvalue and the total cumulative explained variance.

	r_{QQQ}	r_{SPY}	F_r^1
r_{QQQ}	1.00		
r_{SPY}	0.87	1.00	
F_r^1	0.91	0.88	1.00
	OFI_{QQQ}	OFI_{SPY}	F_{OFI}^1
OFI_{QQQ}	1.00		
OFI_{SPY}	0.61	1.00	
F_{OFI}^1	0.82	0.62	1.00

Table 2: Correlation matrices of returns (upper panel) and order flow imbalance (OFI) (lower panel). The upper panel shows the correlation matrix of QQQ ETF returns (r_{QQQ}), SPY ETF returns (r_{SPY}), and the first principal component of returns (F_r^1). The lower panel shows the correlation of the order flow imbalance of QQQ (OFI_{QQQ}) and SPY ETFs (OFI_{SPY}) with the first principal component of OFIs (F_{OFI}^1) (Jan-2008: Jun-2010).

first principal component, while Benzaquen et al. (2017) report a first principal component accounting for 23% of total variation in 5-minute returns across 275 US stocks (NASDAQ and NYSE). Column 3 in Table 1 shows that the second eigenvalue of returns is 1.70, explaining 2.54% of the variance, while column 4 of the same table shows that the third eigenvalue explains 2.21% of the variance. Altogether, the first three principal components of returns account for 36.75% of the total variance, which is higher than the value 27.87% found by Hasbrouck and Seppi (2001).

The first eigenvalue of OFI suggests strong commonality in order flow. The first eigenvalue of OFI shows that 28.27% of the total variance in 1-minute OFI across all stocks can be explained by a single common factor. This is much higher than the 13.5% found by Hasbrouck and Seppi (2001) on a 15-minute interval, although the comparison needs to be taken with caution since the authors used a different variable to summarise the order flow. Column 6 in Table 1 shows that the second eigenvalue of OFI is 1.29, explaining 1.92% of the variance, while the third eigenvalue is 1.21, explaining 1.80% of the variance. Altogether, the first three eigenvalues of OFI account for 32.00% of the total variance, which is smaller than what we found for returns. As a comparison, Hasbrouck and Seppi (2001) found that the first three eigenvalues of their order flow variable accounted for only 20.50% of the total variance.

Figure 6 displays the first three principal components of order flow imbalance. The first principal component, whose loadings across all stocks are positive, clearly represents a 'level' effect. The second principal component is a long/short market neutral portfolio with loadings of opposite sign for pharmaceuticals and semiconductors, while the third principal component also shows clustering of coefficients across sectors.

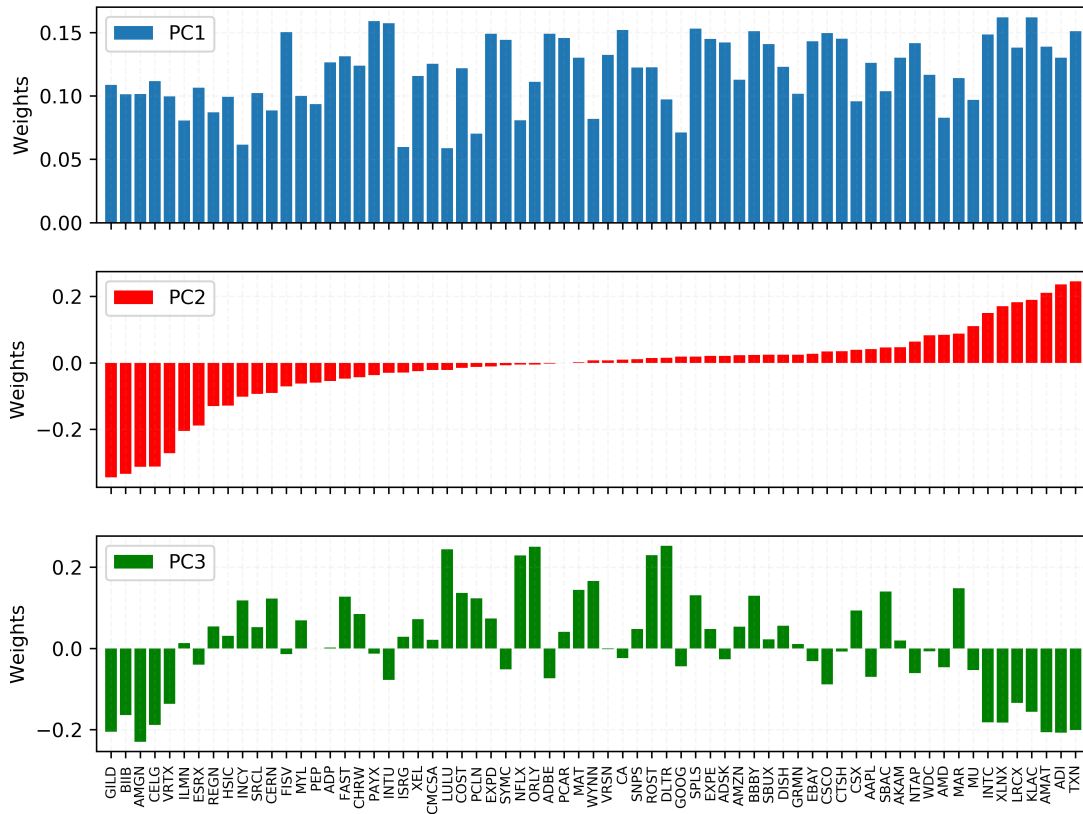


Fig. 6. First, second, and third principal components of 1-minute order flow imbalance (OFI) for NASDAQ-100 (Jan-2008: Jun-2010).

The left panel in Figure 7 shows that the first principal component of order flow imbalance is strongly aligned with that of returns. The corresponding factor processes F_r^1 and F_{OFI}^1 , are also very highly correlated, with a correlation coefficient of 87.26%. The right panel in Figure 7 shows a scatter plot of these factors. The correlation coefficient of the second principal component of returns and the second principal component of OFIs is 53.88%, while the correlation coefficient of the third principal components is 37.16%.

To better interpret the first principal components of returns and OFIs we examine their correlation with the returns and OFIs of the QQQ and SPY ETFs, which track the NASDAQ-100 and the *S&P* 500, respectively. The upper panel of Table 2 shows the correlation matrix of returns, including the returns of the QQQ ETF (r_{QQQ}) the returns of the SPY ETF (r_{SPY}) and the first principal component of returns (F_r^1). We can see from this table that the first principal component of returns is strongly positively correlated with the returns of the overall NASDAQ-100 and *S&P* 500 indices, with correlation coefficients of 91.37% and 88.16%, respectively. Similarly, the lower panel of Table 2 shows the correlation of order flow imbalance for two large ETFs, QQQ (OFI_{QQQ}) and SPY (OFI_{SPY}) with the common factor in order flow (F_{OFI}^1). This common factor is strongly correlated with the overall OFI of the NASDAQ-100, with a correlation coefficient of 82.00%, and with the OFI of the *S&P* 500, with a correlation coefficient of 61.80%. Therefore, one can interpret this common factor as the 'aggregate order flow imbalance'.

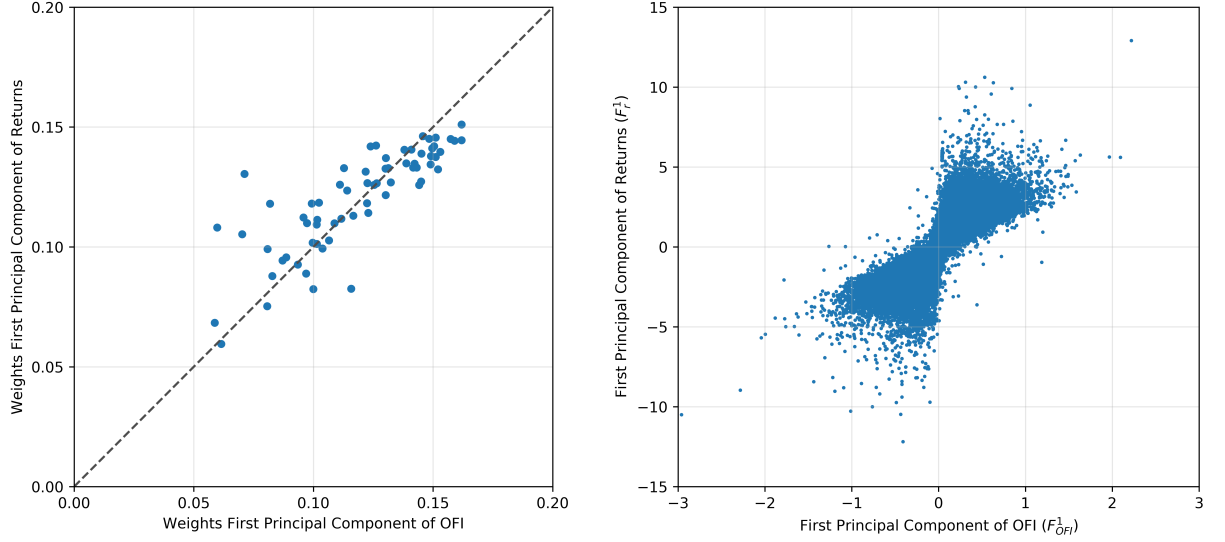


Fig. 7. First principal component (PC) of 1-minute returns (F_r^1) vs. first principal component of 1-minute order flow imbalance (F_{OFI}^1). Left: loadings of the first PC for returns vs first PC for order flow imbalance. Right: time series of $F_{r,t}^1$ vs $F_{OFI,t}^1$ (Jan-2008: Jun-2010). The correlation coefficient is 87%.

3.4. Is there additional explanatory power in cross-impact terms?

In the previous section we found that 32% of the explained variance in returns and 28.27% of the explained variance in OFI is accounted for by their respective first principal components. The question we ask ourselves next is: do we still need cross-impact coefficients once this commonality has been taken into account?

We use a two-step procedure. In the first step, we assume a one-factor structure for the OFIs and we regress each stock's OFI ($ofi_{i,t}$) on the OFIs' common factor ($F_{OFI,t}^1$):

$$ofi_t = \alpha + \gamma F_{OFI,t}^1 + \eta_t. \quad (26)$$

The residuals of this regression, η_t represent the order flow not accounted for by the common factor $F_{OFI,t}^1$.

In a second step, we regress stock returns on the OFI's common factor $F_{OFI,t}^1$, the residuals from the regression in step 1 for stock i ($\eta_{i,t}$) and the residuals from the regression in step 1 for all other stocks $j \neq i$ ($\eta_{j,t}$):

$$r_t = \mu + \gamma F_{OFI,t}^1 + \Phi \eta_t + \xi_t, \quad (27)$$

where Φ is the matrix of market impact coefficients 'purged' by the OFI's common factor $F_{OFI,t}^1$, and η_t is the vector of all stocks' residuals from (26). In this way we are able to disentangle the impact on the returns of stock i due to common risk factors, the stock's own order flow and order flow of all other stocks.

Presence of cross-impact would translate in this setting into off-diagonal coefficients ϕ_{ij} which are statistically significant and of an economically significant magnitude when compared to a stock's idiosyncratic coefficient ϕ_{ii} . Figure 8 shows a heat map of the market impact coefficient matrix Φ , estimated from (27),

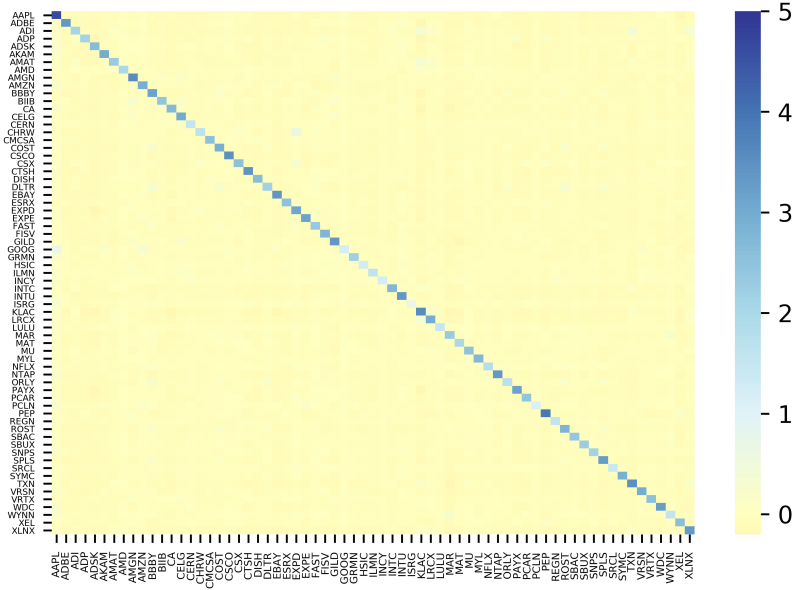


Fig. 8. Heat map of the matrix Φ estimated from Eq. (27) (Jan-2008: Jun-2010).

using heteroskedasticity-consistent Newey-West standard errors. This is quite similar to Figure 4, and once again shows that the off-diagonal terms, representing the impact of a stock's order flow on another stock's returns, once the OFI's first common factor has been taken into account, are much smaller in magnitude compared to the diagonal terms, representing a stock's idiosyncratic impact. Figure 9 presents the relative frequency distributions of the diagonal "idiosyncratic impact" coefficients ϕ_{ii} (upper panel - blue), the off-diagonal "cross-impact" coefficients ϕ_{ij} (upper panel - red), together with the frequency distribution of the coefficients of the OFI's common factor γ_i (lower panel - green). The mean diagonal idiosyncratic impact coefficient ϕ_{ii} is 2.57 with a standard deviation of 0.77, while the mean off-diagonal "cross-impact" coefficient ϕ_{ij} is -0.039 with a standard deviation of 0.06, and the average coefficient for the first principal component γ_i is 2.50, with a standard deviation of 0.39. This means that once we account for the first principal component, off-diagonal terms not only remain very small in magnitude when compared with the diagonal terms, but they are also very small when compared with the coefficients on the first principal component of OFIs.

The diagonal coefficients ϕ_{ii} and all loadings γ_i on the common factor are statistically significant at the 1% significance level. Also, some off-diagonal coefficients (76.25%) remain statistically significant at the 1% significance level. In order to test whether or not we can reject the null hypothesis that the off-diagonal Φ coefficients are jointly equal to zero for any of the 67 regressions, we compare the model in (27), with the nested model in which only a stock's own OFI and the common order flow factor F_{OFI}^1 are taken into account:

$$r_{i,t} = \mu_i + \phi_{ii}\eta_{i,t} + \gamma_i F_{\text{OFI},t}^1 + \xi_{i,t}. \quad (28)$$

Using an F -test we cannot reject the null hypothesis that the off-diagonal coefficients are jointly equal to zero for any of the 67 regressions. However, the increase in explanatory power that one achieves by using model (27) instead of the model (28) with cross-impact with thousands of additional parameters is very

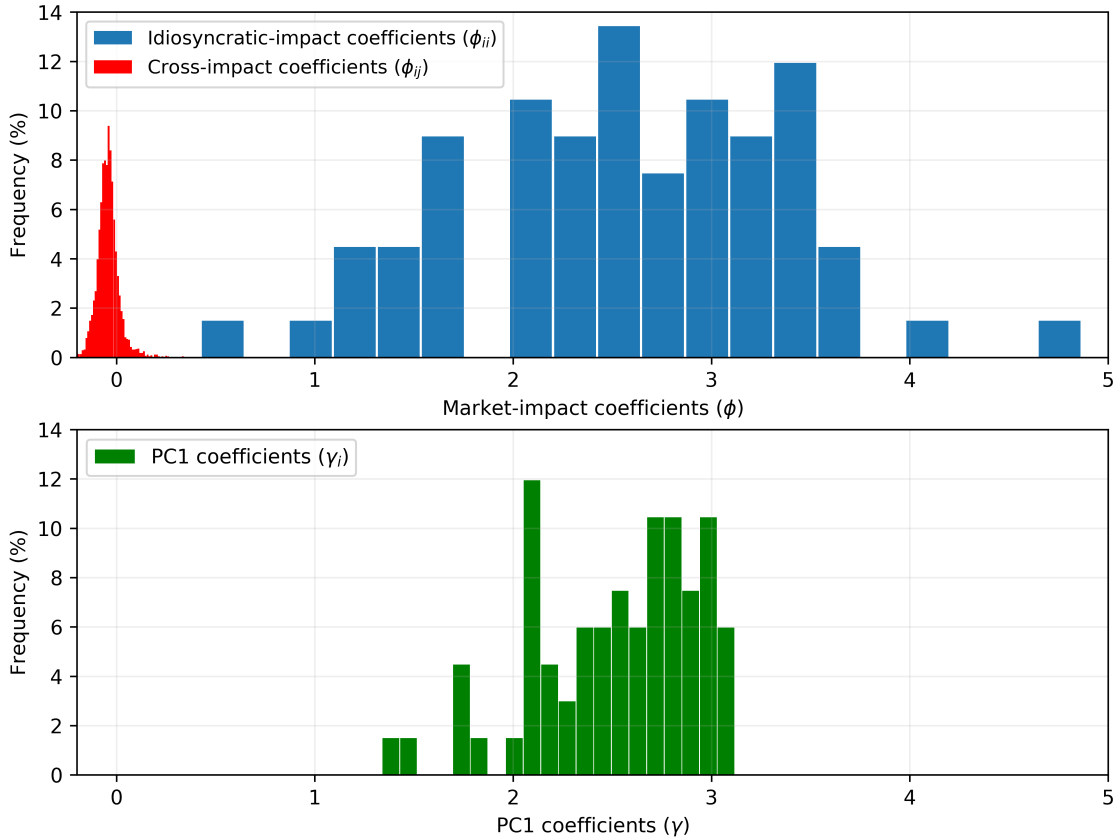


Fig. 9. Estimation results for Eq. (27) (Jan-2008: Jun-2010). Upper panel: distribution of the estimated diagonal idiosyncratic-impact coefficients ϕ_{ii} (blue) and off-diagonal cross-impact coefficients ϕ_{ij} (red). Lower panel: loadings γ_i on common order flow factor (first principal component of OFI).

small. This is shown in Figure 10, where the upper panel presents the histogram of adjusted R-squared when a stock's returns are regressed on a stock's idiosyncratic OFI and the common OFI factor (28). In this case, the average adjusted R-squared is 43.31%, with a standard deviation of 9.8 percentage points. The lower panel of Figure 10 shows the histogram of adjusted R-squared when a stock's returns are regressed on a stock's idiosyncratic OFI, the first principal component of OFIs, and all other stocks' idiosyncratic OFIs (27). We can see that the increase in explanatory power given by adding the order flow of all other stocks is minimal: the average adjusted R-squared is 43.84%, with a standard deviation of 9.8 percentage points. The inset graph in the lower panel of Figure 10 shows the difference in adjusted R-squared between (27) and (28). This graph shows that once a stock's own order flow and the first principal component of OFI have been taken into account, adding the order flow of all other stocks as explanatory variables only increases the adjusted R-squared by 0.5 percentage points, on average.

Thus, despite being jointly statistically significant, the cross-impact coefficients ϕ_{ij} in (27) are not of an economically significant magnitude. The inclusion of all other stocks' OFIs only increases the explanatory power by 0.5 percentage points, on average, but at the same time the number of coefficients to be estimated increases from $67 \times 2 = 134$ to $67^2 = 4,489$. The red histogram in the upper panel of Figure 9 also shows that the interpretation and use of these cross-impact terms is problematic. We can see that once the correlation

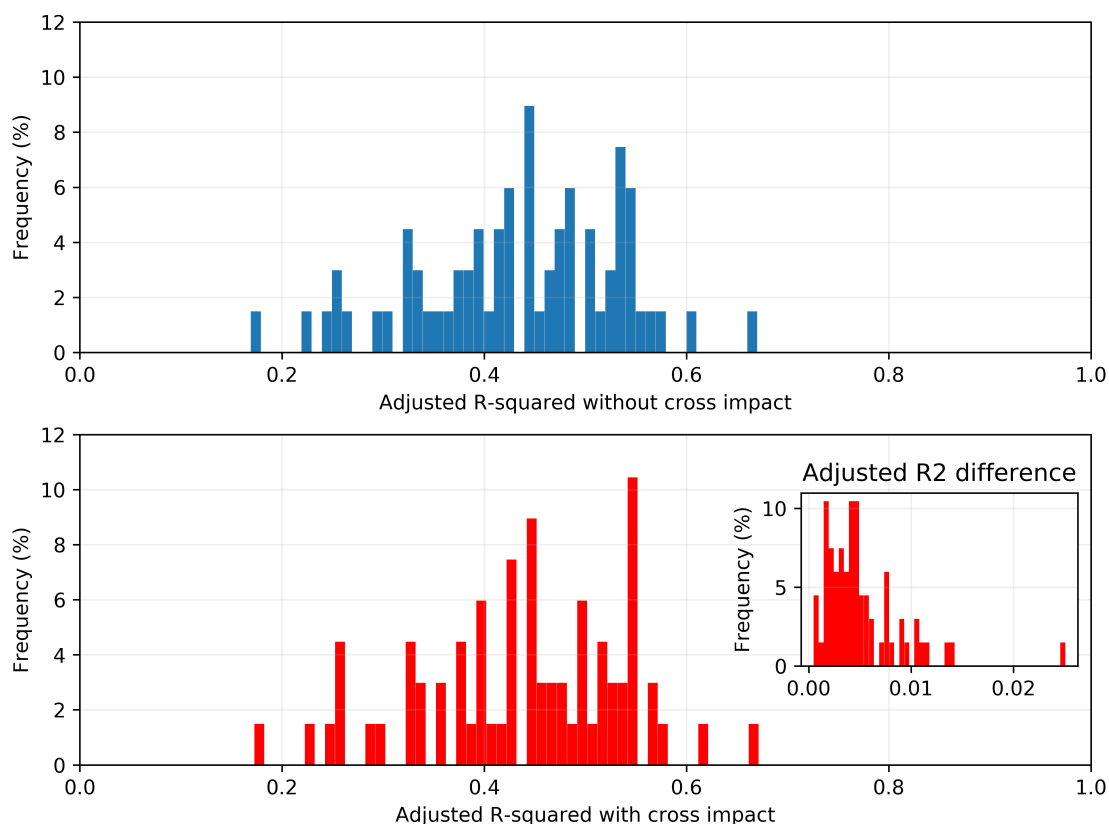


Fig. 10. Histograms of adjusted R-squared from regressions (28) and (27) (Jan-2008: Jun-2010). The upper panel shows the R-squared from Eq. (28), where a stock's returns are regressed on the stock's own residual OFI and the common factor in OFI (no cross-impact). The lower panel shows the R-squared from Eq. (27), where a stock's returns are regressed on the stock's own OFI, the common factor in OFI, and all other stocks' OFIs. The smaller histogram in the lower panel shows the difference in adjusted R-squared between the model with cross-impact in Eq. (27) and the model without cross-impact in Eq. (28).

in order flow is taken into account, most cross-impact coefficients change sign, and become *negative*. Indeed, while the percentage of negative off-diagonal terms from regression (25) is only 23.09%, this percentage increases to 84.46% when we include the first principal component of OFIs in the regression.

The results of this section clearly suggest that the main determinants of execution costs are a stock's own order flow imbalance and the common component in OFI, and that once these two factors have been accounted for, the additional contribution of all other stocks' OFIs is negligible. Thus, accounting for commonality in the order flow seems to be a more effective way to build multi-asset market impact models than introducing 'cross-impact' terms, whose contribution to total impact is very small and whose interpretation is ambiguous. This leads to a more parsimonious approach for modelling multi-asset market impact, in which the number of parameters to be estimated is drastically reduced.

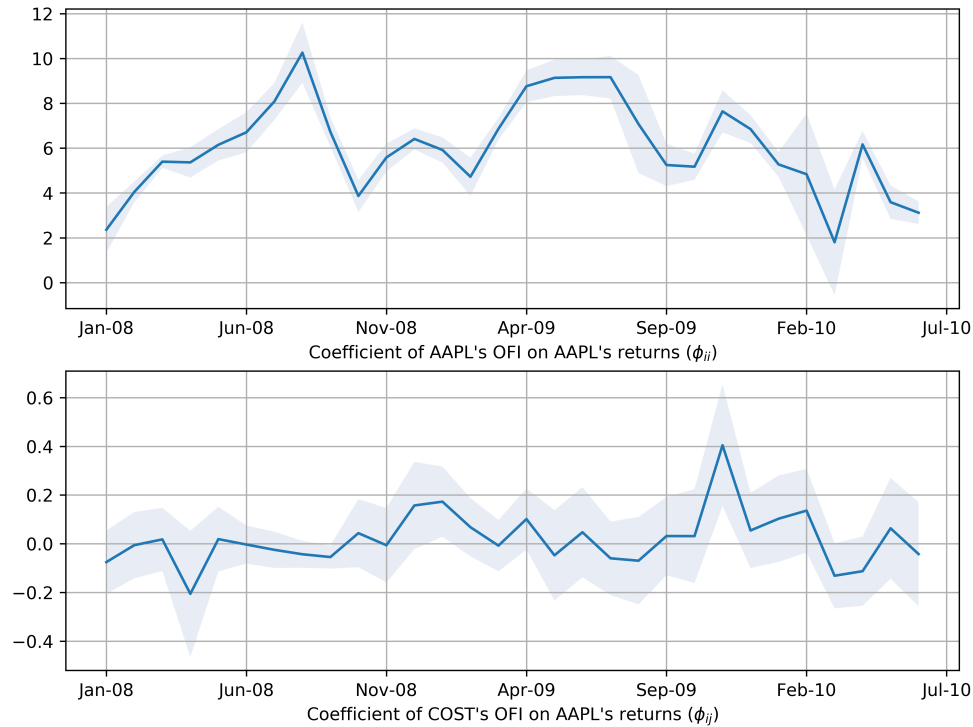


Fig. 11. Evolution of the AAPL's idiosyncratic-impact coefficient ϕ_{ii} (upper panel) and cross-impact coefficient of COST's OFI on AAPL's returns ϕ_{ij} (lower panel) from Eq. (25), computed over a one-month moving window (Jan-2008: Jun-2010). Shaded blue bands represent 99% confidence intervals computed using heteroskedasticity-consistent Newey-West standard errors. The coefficient ϕ_{ij} is unstable and changes sign randomly.

3.5. Instability of cross-impact coefficients

Multivariate market impact models rely on the calibration of the matrix of cross-impact coefficients and its stability over time. Stability of the off-diagonal cross-impact coefficients in Φ is perhaps more important than their statistical significance. Indeed, cross-impact terms whose sign changes over time pose problems for interpretation and use.

In the previous section we found that the vast majority of the off-diagonal terms in Φ become negative once we take into account the commonality in OFIs. To test whether or not these coefficients are stable over time, we divide our sample into 30 monthly sub-samples. For each sub-sample, we perform a PCA on returns and OFIs and repeat the two-step process of (26)-(27).

Figure 11 shows the evolution of the idiosyncratic impact coefficient ϕ_{ii} of Apple Inc. (AAPL) (upper panel) and the "cross-impact" coefficient ϕ_{ij} of Costco Wholesale Corporation (COST) (lower panel) over a one-month moving window with shaded 99% confidence intervals. We can clearly see from the lower panel how the cross-impact coefficient is unstable and changes sign randomly through time.

Similarly, Figure 12 shows the evolution of the idiosyncratic impact coefficient ϕ_{ii} of Apple Inc. (AAPL) (upper panel) and the cross-impact coefficient ϕ_{ij} of O'Reilly Automotive Inc. (ORLY) (lower panel) over a one-month moving window. Once again, the cross-impact coefficient is unstable and changes sign randomly through time.

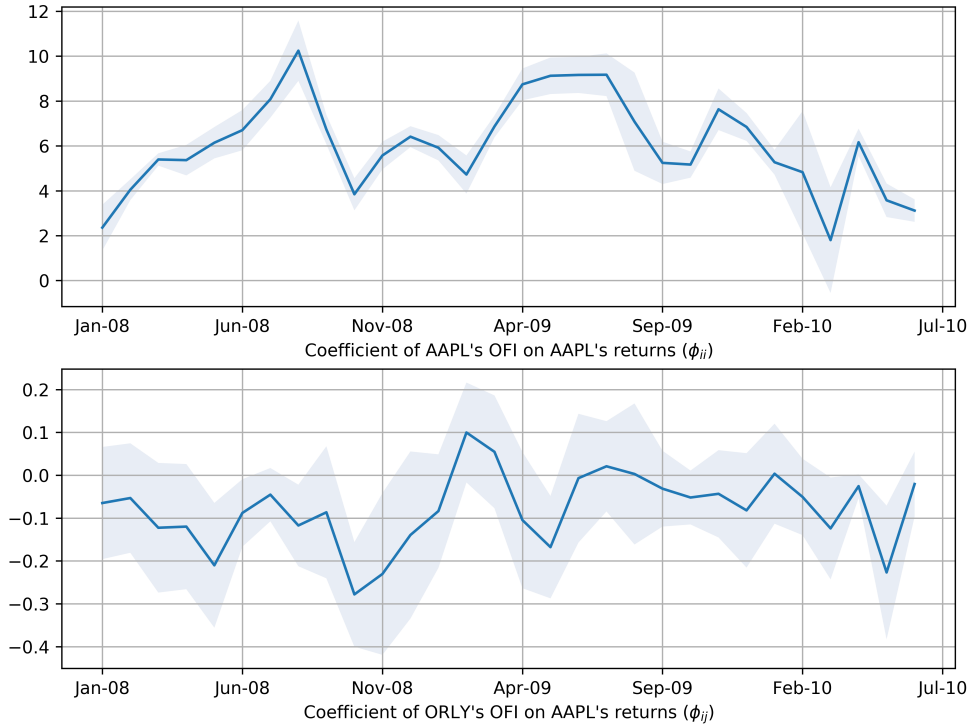


Fig. 12. Evolution of AAPL's idiosyncratic-impact coefficient ϕ_{ii} (upper panel) and the cross-impact coefficient of ORLY's OFI on AAPL's returns ϕ_{ij} (lower panel) from Eq. (25), computed over a one-month moving window (Jan-2008: Jun-2010). Shaded bands represent 99% confidence intervals computed using heteroskedasticity-consistent Newey-West standard errors. The coefficient ϕ_{ij} is unstable and changes sign randomly.

These results clearly show that cross-impact coefficients are not stable over time, even for liquid 'blue chip' stocks. Thus, a cross-impact model that is based on the calibration of Φ under these circumstances may lead to spurious results. Together with the results from the previous section, this suggests that instead of focusing on cross-impact, one should focus on understanding and modelling the correlation in OFIs. One can then simply use the common OFI factor to extend the univariate market impact model without the need for introducing cross-impact terms.

4. Discussion

Recent studies on market impact have introduced the notion of cross-impact to explain the positive covariation between the returns of an asset and the order flow of other assets. We have first shown, through several examples, that one cannot use the existence of such positive covariance terms to infer the existence of cross-impact and that this positive covariance may be explained, without invoking the notion of cross-impact, by simply noticing that order flow imbalances are correlated across stocks, a phenomenon which arises naturally from multi-asset trading strategies.

We have provided empirical evidence to support our explanation: we have shown that returns and order flows are driven by highly correlated common factors, and that this commonality in order flow is the

dominant contribution to the covariance between order flow imbalance and return across assets. Once this common factor in order flow is accounted for, introducing cross-impact terms provides little or no additional explanatory power. We are therefore unable to uncover any empirical justification for including cross-impact mechanisms in market impact model.

Our results demonstrate that the main determinants of execution costs are an asset's own order flow (OFI) and a market order flow factor common across stocks; adding the order flow of all other stocks only increases the explanatory power by 0.5 percentage points on average. Our results also show that once the commonality in order flow has been taken into account, nearly 85% of cross-impact terms become *negative*, and that the sign of these cross-impact terms changes randomly through time. The fact that cross-impact coefficients are small in magnitude and unstable over time poses serious problems for their interpretation and use.

Our findings imply that a simpler and more parsimonious approach which takes into account the commonality in order flow seems to be a more effective way to build multi-asset market impact models than introducing cross-impact coefficients whose statistical significance and interpretation are not clear.

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